

Yijie Deng

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EDUCATION

- PhD of Computer Science** September 2024 - Present
Tandon School of Engineering, New York University (currently visiting Tsinghua University)
GPA: 3.78/4.00
- Master of Data Science and Information Technology** September 2021 - June 2024
Tsinghua-Berkeley Shenzhen Institute, Tsinghua University
GPA: 3.85/4.00
- Bachelor of Computer Science** September 2017 - June 2021
School of Computer Science, Wuhan University
GPA: 3.89/4.00

RESEARCH PUBLICATION

- Yijie Deng**, Lei Han, Tianpeng Lin, Lin Li, Jinzhi Zhang and Lu Fang. *RealLiFe: Real-Time Light Field Reconstruction via Hierarchical Sparse Gradient Descent* accepted by IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) December 2025
- Yijie Deng**, Lei Han, and Lu Fang. *TransGI: Real-Time Dynamic Global Illumination With Object-Centric Neural Transfer Model* accepted by IEEE Transactions on Visualization and Computer Graphics (TVCG) August 2025
- Yijie Deng**, Shuaihang Yuan, and Yi Fang. *AnyImageNav: Any-View Geometry For Precise Last-Meter Image-Goal Navigation* in submission March 2026
- Yijie Deng**, Shuaihang Yuan, Geeta Chandra Raju Bethala, Anthony Tzes, Yu-Shen Liu, and Yi Fang. *Hierarchical Scoring with 3D Gaussian Splatting for Instance Image-Goal Navigation* in submission September 2025
- Yijie Deng**, Shuaihang Yuan, Congcong Wen, Hao Huang, Anthony Tzes, Geeta Chandra Raju Bethala, Yu-Shen Liu, and Yi Fang. *MapBERT: Bitwise Masked Modeling for Real-Time Semantic Mapping Generation* in submission September 2025

PROJECT EXPERIENCE

- Indoor Object Goal Navigation and Image Goal Navigation** January 2025 - February 2026
Role: Algorithm Development
- Developed a real-time indoor map completion algorithm running at 90 FPS, enabling robots to predict semantic information in unexplored areas and discover target objects earlier (75.8% success rate on Gibson).
 - Developed a Gaussian Splatting-based image navigation system that leverages semantic, texture, and geometric cues to guide robots precisely to the target image location (78.4% success rate on HM3D).
 - Developed a 3D foundation model-based image navigation system using image feature matching for faster and more accurate goal localization (82.4% success rate on HM3D, within 5° orientation error).
- Embodied Intelligence and Robot Simulation Practice** July 2025 - January 2026
Role: Algorithm Development
- Conducted VLA and robotic manipulation experiments in Robosuite, Isaac Gym, and related simulation platforms.
 - Built a multi-view data collection pipeline in Robosuite to study VLA sensitivity to viewpoint changes and support generalization analysis.

- Developed basic grasping environments in Isaac Gym and PyBullet, integrating semantic segmentation and target localization for object-specific grasping.

Real-Time Cloud Rendering April 2023 - March 2024
Role: Algorithm Development

- Developed a cloud-assisted rendering system for high-quality real-time graphics on low-end devices.
- Proposed a compact object-centric radiance transfer representation with improvements in compression and rendering decomposition.
- Implemented the system in NVIDIA Falcor, achieving 15 FPS at 1920×1080 with image quality comparable to ray tracing.

Real-Time Light Field Reconstruction with Hierarchical Sparse Gradients October 2022 - April 2023
Role: Algorithm Development

- Developed a real-time light field video generation method for 3D displays and VR devices.
- Used sparse light field gradients to improve efficiency while maintaining visual quality.
- Achieved the fastest reported light field generation speed, 400× faster than DeepView and 2 dB higher PSNR than IBRNet, MVSNerf, and ENerf.

Development of a GLSL/C++ Ray Tracing Renderer June 2023 - August 2023
Role: Software Design and Development

- Developed a physically based path tracing renderer using GLSL/C++.
- Implemented Disney BSDF and basic support for loading Blender files.
- Enabled interactive rendering and improved the usability of the PBR rendering pipeline.

Virtual Scene, Dataset Generation, and VR Interaction Development with Unreal Engine and Unity
 January 2021 - June 2021
Role: Software Design and Development

- Developed virtual scene and interaction systems using Unreal Engine and Unity for both dataset generation and VR applications.
- Built a virtual pedestrian dataset platform with scene simulation, behavior control, and annotations including G-buffer and trajectories.
- Developed an educational VR escape-room game themed around ancient Egypt, including puzzle design, player control, and headset interaction.

INTERNSHIP

Centific (Microsoft) Internship,	
Position: Algorithm developer of generative video editing	December 2025 - April 2026
Huawei Technologies Co Ltd Internship,	
Position: Algorithm developer of light field and real-time rendering	October 2022 - March 2024
Eon Reality Internship,	
Position: Game designer and programmer of virtual reality	June 2020 - August 2020

GRANTS AND AWARDS

Outstanding Graduate in School of Computer Science	2021
National English Competition for College Students, Second Prize of Hubei Province	2019
National Encouragement Scholarship (5000 CNY * 4 years)	2017-2021
"TianYuanDiKe" Special Scholarship (8000 CNY *1 year)	2018
First-class Scholarship for Excellent Students (3000 CNY * 4 years)	2017-2021
Intermediate Software Design Qualification	2019
National Mathematics Competition for College Students, Second Prize of Hubei Province	2018
Nationally Televised CCTV New Hope English Speech Competition, First Prize of Hubei Province	2018